

Baud Rate Detection Tool

Instruction Manual

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Part I Overview

Baud rate detection tool is used to realize in the case of unknown CAN bus baud rate, through the USB-CAN adapter in accordance with a certain law scanning bus baud rate, and ultimately determine the baud rate of the bus operation of the utility tool.

This tool currently supports USB-CAN and CANalyst-II series adapters of various types (CANalyst-II analyzer select USB-CAN2.0 when choosing the device model)!

To detect the bus baud rate, the following conditions should be met:

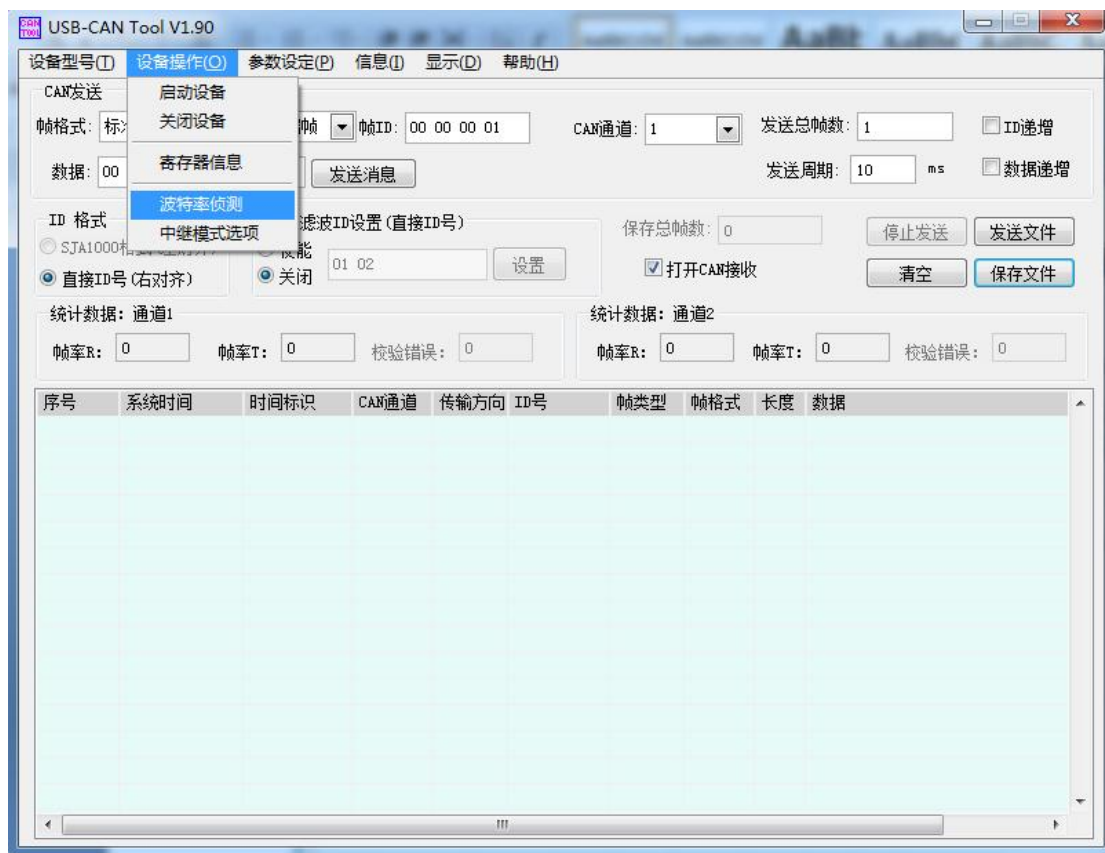
1. If more than one node exists on the CAN bus, these nodes must be operating at the same baud rate. That is, the bus is working properly;
- 2, At least one node on the CAN bus is in working condition, i.e. this node has to continuously send CAN messages to the bus during the baud rate detection.

Part II Operating Procedures

2.1 Operation Procedure

Before baud rate detection, please make sure that the USB-CAN adapter has been correctly connected to the CAN bus to be detected, plugged into the USB port of the computer, and the hardware driver has been successfully installed; the USB-CAN Tool software has been installed in the computer.

First, double-click the desktop "USB_CAN TOOL" software shortcut to open the USB_CAN TOOL software.



Open the baud rate detection tool, you will see the interface as shown in the following figure, 1, select the device type, device index and CAN channel number; 2, click on the "start device" to open the device; 3, start fast detection or range detection.

If you need to stop the detection process, please click the "Stop" button at the bottom right corner of the interface!



2.2 List Items

In the software interface, the list display area, lists the detailed parameters of the baud rate, explained as follows:

- 1、Baud rate: the actual value of the baud rate of the detection
- 2、Status: Display the current detection status
- 3、Timing0(BTR0): Compatible with the SJA1000 controller's BTR0 register value
- 4、Timing0(BTR1): Compatible with SJA1000 controller's BTR1 register value
- 5、Synchronous jump width: the baud rate parameter of the adapter used by our company, there is a corresponding option in the parameter setting of USBCAN TOOL in the random software.
- 6、Pre-split Frequency: the baud rate parameter of the adapter used by our company, there is a corresponding option in the parameter setting of USBCAN TOOL in the random software.
- 7、Sampling Point: The baud rate parameter of the adapter used by our company has corresponding options in the parameter setting of USBCAN TOOL in the random software.
- 8、Synchronization section: the baud rate parameter of the adapter used by our company, there is a corresponding option in the parameter setting of USBCAN TOOL

in the random software.

9、Propagation time segment: the baud rate parameter of the adapter used by our company, there is a corresponding option in the parameter setting of USBCAN TOOL in the random software.

10、Phase buffer segment 1: the baud rate parameter of the adapter used by our company, there is a corresponding option in the parameter setting of USBCAN TOOL in the random software.

11、Phase buffer segment 2: the baud rate parameter of the adapter used by our company, there is a corresponding option in the parameter setting of USBCAN TOOL of the random software.

Note: For the detailed introduction of the baud rate parameters of the adapter used by our company, please refer to the relevant instruction manual. After successfully detecting the bus baud rate, you can use the detailed parameters of the detected baud rate to set up the CAN adapter parameters in the USB CAN TOOL or during the secondary development to achieve the effect of matching the bus baud rate!

2.3 General Baud Rate Detection

Detect commonly used baud rate, such as 10kbps, 50kbps, 125kbps, 250kbps, 500kbps, 1000kbps, etc. It is faster.

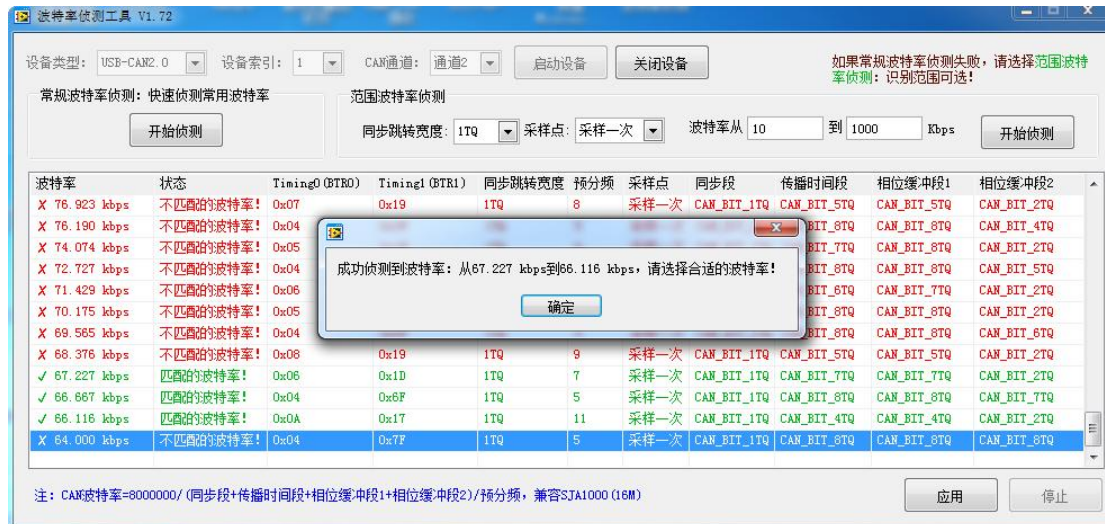
2.4 Range Baud Rate Detection

Specify the baud rate detection range, within the specified range, the baud rate will be detected in detail, applicable to the situation where the regular baud rate can not be detected, slow speed, but high success rate!

2.5 Detection Result

Generally, when the baud rate is successfully detected, the result will be displayed. The result is generally a range of baud rates (because baud rates are

allowed to have a 1% error, i.e. as long as the baud rate is close, you can communicate normally), as shown in the figure below. At this point, it can be converted, the center of this baud rate range that is the most accurate baud rate, the figure is 66.667kbps.

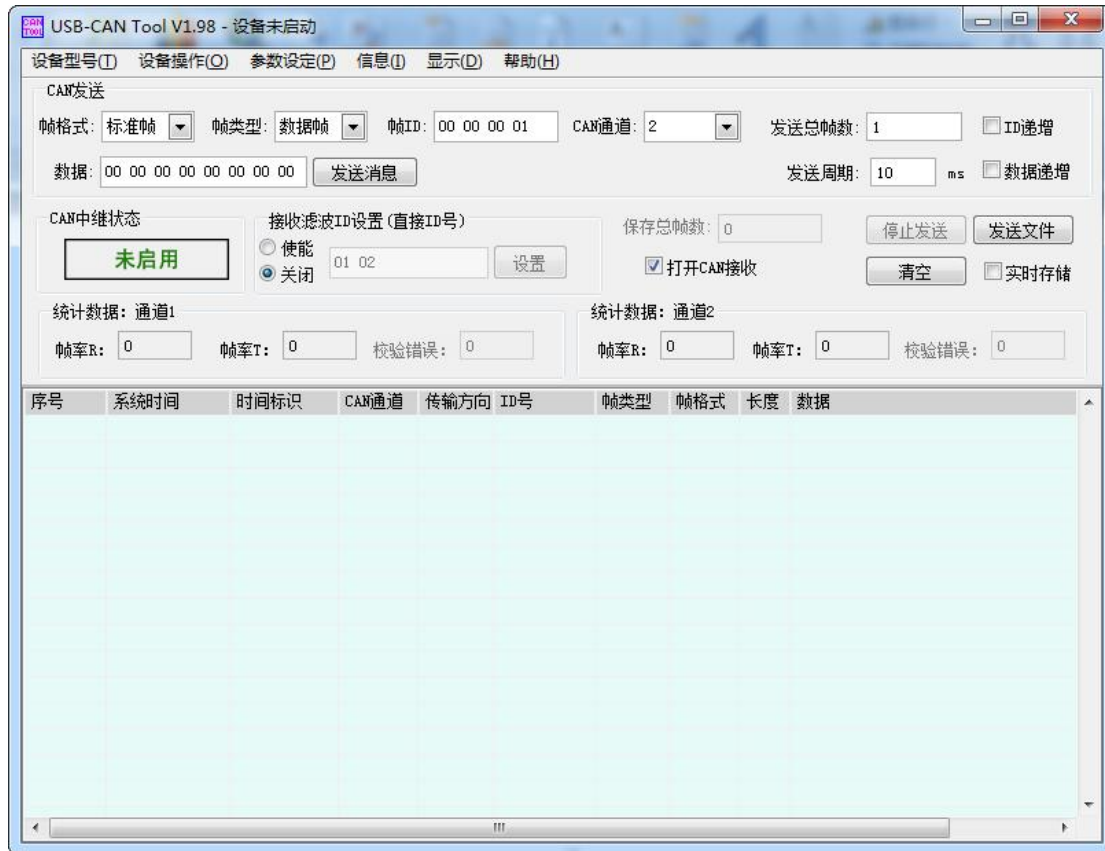


2.6 Application of Detection Result

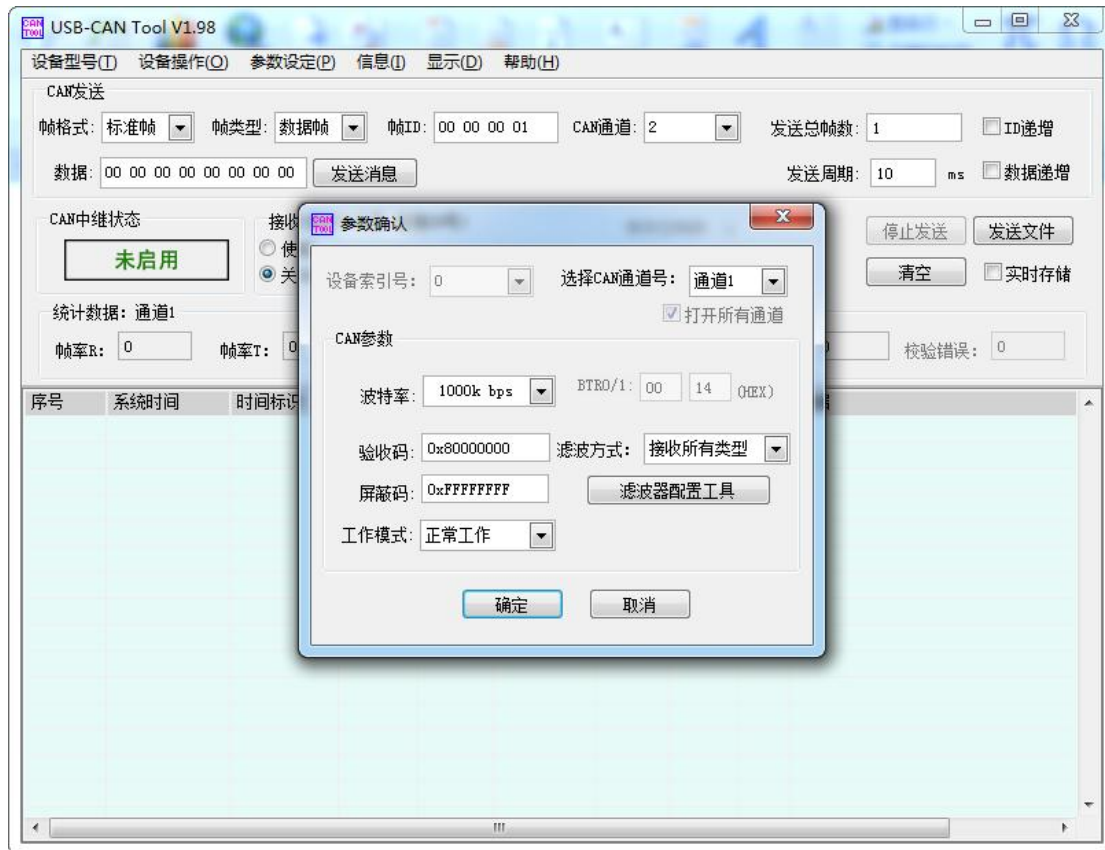
Generally, after successfully detecting the baud rate, the detection result will be displayed, and we need to apply the result of the frame detection to the USB_CAN TOOL software in order to monitor the bus data. As the above example, the best baud rate of 66.667kbps is detected and the corresponding parameter list shows:

- 1、Timing0(BTR0): 0x04
- 2、Timing0(BTR1): 0x6F

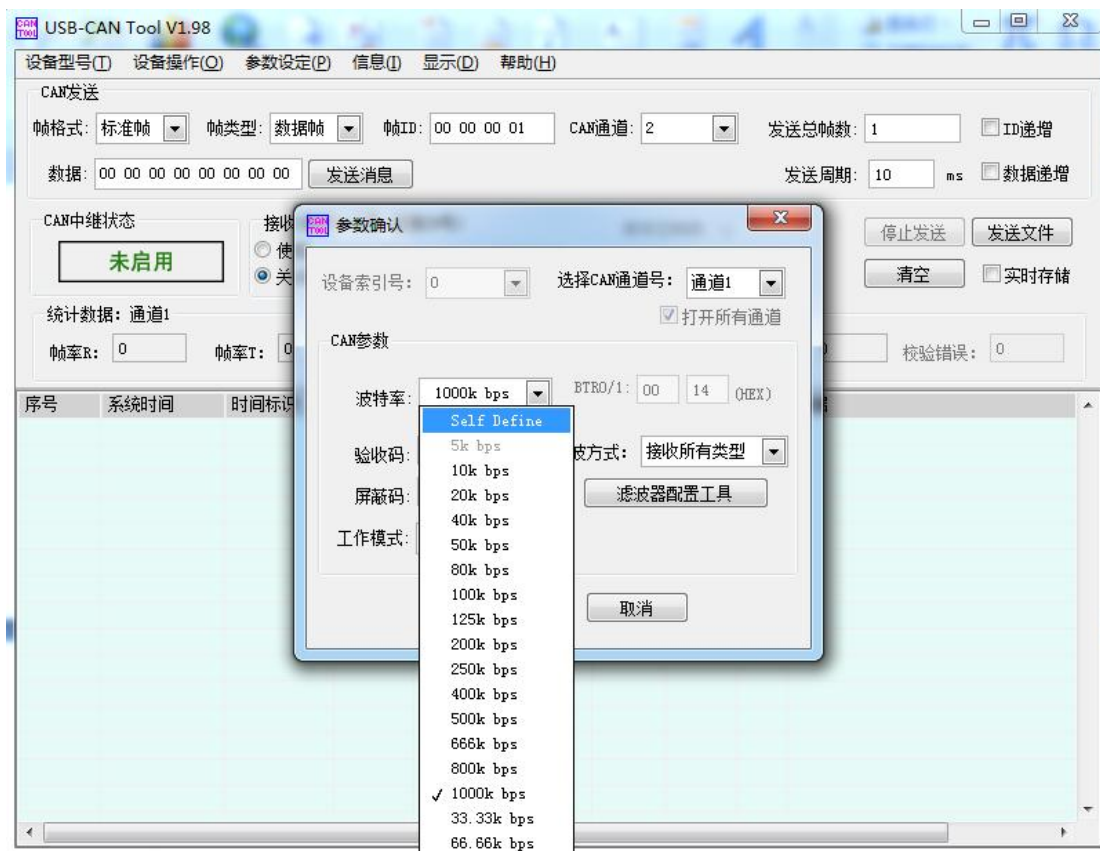
① Start the USB_CAN TOOL software.



③ Configuration Parameters



④ Customized Baud Rate



⑤ Fill in the Detection Results

